Enhancing Leaf Nitrogen Estimation From Satellite Multispectral Imagery via Hyperspectral Super-Resolution and Structural Correction

Tinghan Wang, Liangsheng Shi, and Chenye Su

***Abstract*—Leaf nitrogen content (LNC) is a key regulator of photosynthesis and crop (CRO) productivity, and its timely monitoring underpins both precision nitrogen management and large-scale ecosystem assessment. Remote sensing offers non-destructive, spatially explicit monitoring of LNC, but retrieval accuracy is degraded by canopy structural effects and the limited spectral resolution of multispectral (MS) satellite sensors. Here, we develop an integrated framework that combines spectral super-resolution (SSR) with canopy structure correction to improve satellite-based LNC estimation. Sentinel-2 MS reflectance is first reconstructed into full hyperspectral spectra using a denoising diffusion probabilistic model (DDPM). Directional area scattering factor (DASF) is then computed from the reconstructed spectra to derive the canopy scattering coefficient (CSC), which normalizes structural effects and serves as input to a random forest regression (RFR) model for LNC estimation. On simulated data, the DDPM achieves high reconstruction fidelity (PSNR = 41.49 dB and SAM = 0.9989). Using CSC derived from DDPM outputs improves LNC estimation compared with MS bidirectional reflectance factor (BRF) alone (R^2 = 0.579, RMSE = 0.431%, and RRMSE = 23.58%; RMSE reduced by 14.8% and R^2 increased by 0.134). Even when only RGB bands are available, the super-resolution + CSC pipeline increases LNC retrieval accuracy (R^2 + 0.14), and the addition of an NIR band further enhances both reconstruction quality and LNC estimation. This study shows that hyperspectral super-resolution coupled with structural normalization can unlock more accurate, scalable LNC monitoring from existing satellite archives, supporting operational nitrogen management and ecosystem monitoring in data-limited regions.**

***Index Terms*—Canopy structure correction, deep learning, leaf nitrogen content (LNC), multispectral (MS) satellite data, spectral super-resolution (SSR).**

I. Introduction

LEAF nitrogen content (LNC) is a key biochemical indicator in terrestrial ecosystems, closely linked to plant physiology, photosynthesis, and carbon sequestration [1], [2]. As an essential component of chlorophyll and photosynthetic enzymes, nitrogen allocation in leaves directly influences vegetation productivity and ecosystem function [3]. In agriculture, maintaining optimal LNC is crucial for maximizing crop (CRO) yields, while excessive fertilization raises environmental risks such as eutrophication and greenhouse gas emissions [4], [5]. As a result, accurately assessing LNC is essential for understanding plant photosynthesis, advancing carbon neutrality research, and guiding informed agricultural decisions. However, traditional methods of LNC measurement rely on destructive sampling and laboratory analysis, which are time-consuming and limited in spatial coverage, making large-scale real-time monitoring of LNC a challenging task. The utilization of satellite remote sensing for the real-time and precise monitoring of LNC has become increasingly important [6], [7].

Remote sensing methods for monitoring LNC have advanced rapidly and can be broadly grouped into empirical, physical, and hybrid approaches. Empirical models exploit statistical or machine learning relationships between LNC and bidirectional reflectance factor (BRF) or vegetation indices [8], [9]. Physical models, grounded in radiative transfer models (RTMs), simulate the interactions between leaf optical properties and biochemical components, enabling the retrieval of leaf biochemical traits by minimizing a cost function [10], [11]. Because most RTMs do not explicitly represent nitrogen, LNC is usually inferred indirectly from retrieved chlorophyll, dry matter, or protein content [12]. Hybrid methods combine simulated spectra from RTMs with observational data to train machine learning models, thereby merging data-driven flexibility with physically based constraints and improving model robustness and transferability [13], [14].

Despite these advances, accurate canopy-scale LNC retrieval remains difficult because canopy structure strongly affects the radiative transfer process and thus the sensitivity of reflectance to leaf optical properties [15], [16], thereby introducing substantial uncertainties in nitrogen estimation. Several studies have been proposed to mitigate the influence of canopy structure on canopy BRF in the estimation of LNC or Cab. The first method is to explore different vegetation index combinations and multiangle vegetation indices to reduce sensitivity to canopy structure [17], [18], [19]. Another treats canopy reflectance as the combined expression of multiple

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traits, including leaf physiological attributes (e.g., LNC) and structural variables (e.g., LAI and biomass), and uses spectral decomposition to separate their contributions [20]. However, both methods mitigate canopy effects only in post-analysis, and the vegetation index method exhibits poor portability, while the spectral decomposition method requires additional prior trait information for practical applications. Furthermore, Knyazikhin et al. [21] proposed a method based on the spectral invariant theory, which utilizes the ratio of BRF and directional area scattering factor (DASF)—referred to as the canopy scattering coefficient (CSC)—to mitigate the impact of canopy structure in the estimation of LNC. Subsequently, Wang et al. [22] successfully applied CSC to correct canopy effects, improving the accuracy of forest LNC estimation, while this method was limited to hyperspectral data.

The DASF is defined as the ratio of leaf area forming the canopy boundary along a given direction to the total leaf area [23]. Its calculation requires specific hyperspectral wavebands (710–790 nm), whereas most satellite imagery provides only multispectral (MS) data and even lacks wavebands within this range, such as Landsat-7/8. This limitation renders the computation of DASF infeasible using common satellite platforms. Pan et al. [24] assumed that the DASF of CROs changes periodically and estimated the DASF of wheat at the UAV scale using NDVI, RAGDD, and band 570. Several studies have proposed near-infrared reflectance of vegetation (NIRv) as an approximate surrogate for DASF [25], [26], [27]. NIRv, defined as the product of NIR reflectance and NDVI, is sensitive to canopy structure and background and has been used to reduce structural effects in chlorophyll and LNC retrievals [14], [27]. However, the accuracy of these estimations may still be limited by the approximations inherent in using NIRv. Given that DASF directly characterizes the fraction of leaf area forming the canopy boundary in a given direction, it inherently captures structural effects more precisely than NIRv. Therefore, accurate DASF estimates from satellite data would provide a more physically consistent way to correct canopy structure and enable more reliable retrieval of biochemical traits such as LNC and chlorophyll.

Spectral super-resolution (SSR) methods leverage machine learning, statistical modeling, or physical constraints to reconstruct high-resolution hyperspectral reflectance from MS observations [28], [29], [30], [31]. Recent work has proposed sophisticated architectures such as gradual spatial–spectral fusion networks [32], iterative guiding networks with deformable fusion [33], and diffusion-based generative models that progressively refine spectral trends to tackle the ill-posed SSR problem [34], [35]. Reconstructed hyperspectral information can compensate for missing spectral features and improve downstream retrievals. Several studies have begun to explore SSR specifically for plant physiological parameter estimation [36], [37]. For example, Lao et al. [38] extended UAV MS data to hyperspectral reflectance using constrained spectral matching, enabling accurate retrieval of leaf nitrogen and phosphorus even when key nitrogen-sensitive bands were absent [38]. In principle, reconstructing full-range hyperspectral spectra from satellite MS data would simultaneously overcome the DASF band limitation and provide richer spectral detail for LNC estimation. However, most existing SSR studies target tens of bands within limited wavelength ranges, and it remains unclear whether high-quality 400–2500-nm spectra can be reconstructed from MS satellites and how such reconstructions influence DASF computation and operational LNC retrieval.

To address the above issues, this study aims to reconstruct hyperspectral data from MS data to compute DASF to correct canopy structural effects on canopy BRF, thereby improving the LNC estimation method based on satellite MS imagery. An overview of the proposed framework is shown in Fig. 1. We aim to address the following questions.

1) Can broadband MS data be well reconstructed into hyperspectral data?

2) Can reconstructed hyperspectral data improve LNC estimation?

3) Can the DASF calculated based on reconstructed hyperspectral data better correct canopy structural effects?

II. Materials and Methods

A. Ground Sampling Data

In this study, we utilized LNC samples from the National Ecological Observatory Network (NEON) dataset [62]. We used LNC samples collected during 2017–2022 from different vegetation types, including cultivated CROs, deciduous broadleaf forest (DF), evergreen needleleaf forest (EF), grassland (GRA), mixed forest (MF), shrubland (SHR), and wetland (WET). This publicly available dataset contains the quality-controlled, native sampling resolution data from NEON's measurement of foliar traits in sunlit, vegetation samples. Trait measurements include leaf mass per area, leaf water content, chlorophyll, carbon and nitrogen concentrations, stable isotopes, major and minor elements, and lignin. Samples were oven-dried at 60 °C–70 °C to constant mass and finely ground. Total nitrogen concentration was determined using elemental analysis with a CHN elemental analyzer, calibrated against standard reference materials to ensure data accuracy and consistency [62].

B. Satellite Data

Sentinel-2, as one of the most widely used satellites for remote sensing monitoring, offers high spatial resolution and a high revisit frequency (five days). It is also the only satellite that provides three bands in the red-edge region, which is highly effective for vegetation monitoring. Therefore, in this study, we used Sentinel-2 MS data to reconstruct hyperspectral data and estimate LNC. To obtain canopy BRF data corresponding to ground-measured LNC, we downloaded Sentinel-2 MS imagery from the Copernicus Data Hub during 2017–2022. Only images with a temporal difference of no more than six days between satellite observations and ground measurements were selected, with a cloud cover threshold set to 10%. We used only 12 out of the 13 wavebands of Sentinel-2 images, as Band 10 related to cirrus clouds was not available. To match the plot size of ground sampling (20×20 m), we extracted a 2×2 pixel region centered on each ground sampling site and calculated the average BRF values. Finally, we obtained the number of BRF-ground LNC sample pairs and the LNC range for each vegetation type, as shown in Table I.

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TABLE I
SAMPLE NUMBER AND LNC RANGE FOR EACH VEGETATION TYPE

nlcdClass	DF	MF	EF	GRA	WET	CRO	SHR
sample number	323	48	203	94	75	12	55
LNC range(%)	0.97 - 4.34	0.76 - 2.94	0.45 - 4.75	0.57 - 3.16	0.45 - 3.10	0.60 - 3.86	0.75 - 4.40

TABLE II
INPUT RANGES OF MODEL PARAMETERS FOR PROSAIL-5B INVERSION AND SIMULATIONS. AVE AND STD INDICATE THE AVERAGE AND STANDARD DEVIATION OF DATA WITH THE NORMAL DISTRIBUTION

Parameter	Unit	Range	Ave	Std
Leaf structure parameter (Nstruct)	/	1-3	1.5	0.5
Leaf chlorophyll content (Chl)	$\mu g/cm^2$	5-100	45	20
Leaf brown pigment content (Cbp)	$\mu g/cm^2$	0-3	1	0.4
Leaf carotenoid content (Car)	$\mu g/cm^2$	0-15	7	3
Leaf water content (Cw)	g/cm^2	0.02	/	/
Leaf dry matter content (Cm)	g/cm^2	0.001-0.05	0.025	0.02
Leaf area index (LAI)	m^2/m^2	0-9	4.5	4
Average leaf angle (ALA)	°	0	50	7
Hot-spot size parameter (Hspot)	m/m	0.5/LAI	/	/
Viewing zenith angle (VZA)	°	0	/	/
Solar zenith angle (SZA)	°	15-50	35	10
Relative azimuth angle (RAA)	°	0-180	80	50
Soil factor (Psoil)	/	0-1	0.5	0.25

C. Simulated and Ground-Based Canopy BRF Data

Simulated data are widely applied in plant trait retrieval, such as examining lookup table techniques and hybrid physical–empirical methods [39], [40]. By configuring different ranges and combinations of input parameters, synthetic datasets can cover a wide range of biochemical components and canopy properties. The PROSAIL model couples the canopy model SAIL [41] with the leaf model PROSPECT [42] to simulate vegetation canopy spectra and biochemical parameters. It is currently one of the most widely used RTMs. Therefore, we decided to use the PROSAIL-5B model [10] to generate a simulated dataset to train the SSR model. We determined the range of model parameters based on corresponding chlorophyll values and published studies (Table II) [14]. The synthetic PROSAIL dataset is not directly used to build LNC estimation models; instead, it serves exclusively as training input for the SSR models. The purpose of these simulations was to provide a sufficiently diverse set of spectral–structural variations rather than to preserve strict biochemical sensitivity, and we will validate its applicability on a limited set of ground data. Furthermore, LNC and chlorophyll content are strongly correlated across vegetation types, meaning that PROSAIL variations in chlorophyll content and canopy structure indirectly span a major proportion of nitrogen-related spectral variability. Although the PROSAIL parameter space used in this study follows the ranges commonly adopted in chlorophyll and canopy structure-oriented research, it is sufficient to cover the range of spectral variations required for training robust SSR models capable of downstream LNC estimation. To obtain the input data for the super-resolution model, we resampled the simulations to broadband BRF spectra for each waveband in Sentinel-2 imagery using the relative spectral response functions as follows [62]:

$$\mathbf{BRF} = \frac{\sum_{\lambda=1}^{\lambda=N} \beta(\lambda) * \mathrm{BRF}'(\lambda)}{\sum_{\lambda=1}^{\lambda=N} \beta(\lambda)} \tag{1}$$

where BRF' (λ) is the original hyperspectral BRF simulated by the PROSAIL-5B model and $\beta(\lambda)$ is the weight of the relative spectral response functions. Finally, we generated a dataset consisting of 10 000 spectra and corresponding Sentinel-2 broadband BRF spectra. The dataset was divided into training, validation, and testing sets in an 8:1:1 ratio.

The 648 field canopy reflectance data points were acquired using the ASD FieldSpec4 spectroradiometer. Canopy reflectance data were obtained from two sources: artificially field experimental plots and naturally farmland. The artificially controlled experiment was conducted for two consecutive years (2022 and 2023) in Hengsha Township, Chongming District, Shanghai, China (31.34°N, 121.84°E). Nanjing 46 rice variety was planted in 12 plots (11 × 5.5 m), and four gradient nitrogen application treatments were applied. The natural field was selected from farmer-managed plots surrounding the control plots. Sampling was conducted from mid-July to early October each year. Measurements were conducted at approximately 10 cm above the rice canopy, ensuring a nadir viewing geometry and minimizing background soil interference. Each spectrum was averaged over ten scans to reduce sensor noise. These canopy reflectance measurements and resampled MS data served as the reference dataset for evaluating the hyperspectral super-resolution reconstruction.

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D. SSR Methods

To reconstruct broadband MS BRF (12 Sentinel-2 bands) into full-spectrum hyperspectral BRF (400–2500 nm), we adopted the U-Net IMAGEN conditional denoising diffusion probabilistic model (DDPM) model [43] as our SSR framework. We used the simulated hyperspectral BRF as the target data and the broadband BRF resampled using the Sentinel-2 spectral response functions as the input.

The DDPM model is a diffusion probabilistic model. The goal of this model is to reverse the Gaussian diffusion process by iteratively recovering the signal from noise through a reverse Markov chain conditioned on the original data. It defines the Markov chain, in which Gaussian noise is gradually added to the data as in a diffusion process, progressively transforming the original data distribution into a simple Gaussian distribution. A deep neural network is then trained to learn the reverse diffusion process, generating the desired data samples by denoising [44]. This denoising-based generation mechanism allows the model to learn complex data distributions by restoring the signal from noise over multiple iterations, making DDPM a powerful framework for data synthesis and super-resolution tasks.

The diffusion model involves a forward process and a reverse process. The forward process q gradually adds Gaussian noise to a high-resolution BRF y_0 over T iterations

$$q\left(y_t|y_{t-1}\right) = \mathcal{N}\left(y_t; \sqrt{1-\beta_t}y_{t-1}, \beta_t \mathbf{I}\right) \tag{2}$$

where y_t is the noisy BRF data at time step t. β_t is a hyperparameter controlling the noise level. $\mathbf{I}$ represents the identity matrix, indicating that the Gaussian noise added during the diffusion process is isotropic.

The reverse process aims to denoise the data, starting from Gaussian noise y_t

$$p_\theta\left(y_{t-1}|y_t, x\right) = \mathcal{N}\left(y_{t-1}; \mu_\theta\left(x, y_t, t\right), \sigma_\theta\left(x, y_t, t\right)\right) \tag{3}$$

where μ_θ and σ_θ are neural network parameterizations that predict the mean and variance of the denoised data, respectively; and x is the MS BRF condition. The conditional denoising diffusion model is trained to minimize the following objective:

$$L = \mathbb{E}_{(x,y)}\mathbb{E}_{\epsilon\sim\mathcal{N}(0,I),t}\left\|\epsilon_\theta\left(x, y_t, t\right) - \epsilon\right\|^2 \tag{4}$$

where $\epsilon \sim N(0, I)$; (x, y) is sampled from the training dataset; and $\epsilon_\theta(x,y_t,t)$ is the noise predicted by the neural network given x, y_t, and t and parameterized by θ. This allows us to predict the noise at each step to compute $\mu_\theta(x,y_t,t)$, thereby progressively generating the high-resolution BRF through the reverse diffusion process.

The core of our DDPM is the U-Net denoiser model ϵ_θ. The network takes as input a concatenation of the noisy hyperspectral BRF y_t (with one channel) and the conditional MS BRF x (with one channel) along the channel dimension as input, obtaining an input with a channel size of 2. The noise predictor at each step of the denoising process is implemented using a shared U-Net architecture. To fully leverage the capability and accelerate computation, we retained only the BRF within the 400–2335-nm range. This allowed the 1-D spectral vector to be reshaped into a 44×44 2-D matrix, enabling more effective learning through the convolutional architecture. The U-Net follows an encoder–decoder structure with residual blocks and skip connections. The encoder starts with an initial convolution projecting the input to 64 channels, followed by three downsampling stages with channel multipliers [1, 2, 2], resulting in feature channels of [64, 128, 128]. Each stage contains two attention blocks that integrate noise level embedding via positional encoding and MLP, employ group normalization, and utilize self-attention at the 22 resolution levels. Spatial downsampling is achieved between stages using a downsample module. The bottleneck is processed by middle residual blocks with self-attention. The decoder symmetrically consists of three upsampling stages using upsample modules, with each stage containing attention blocks that concatenate features from encoder skip connections. All residual blocks employ a dropout rate of 0.2. The final convolutional layer projects the 64-channel features to one output channel, matching the target hyperspectral BRF dimension. The model was trained for a total of 100 000 iterations. We used the Adam optimizer, and the learning rate is initially set to 10^{-4}. The diffusion process was configured with $T = 2000$ timesteps, employing a linear noise schedule where the variance β_t increases from $\beta_1 = 10^{-6}$ to $\beta_T = 10^{-2}$.

We compared our DDPM-based model with three representative models: 1) a baseline U-Net model [45] that shares the same architecture but is trained using a BRF-based mse loss function (CNN-SSR); 2) a generative adversarial network (GAN-SSR) [46]; and 3) a physics-constrained model (Physics-SSR) that is the U-Net model with spectral smoothness and sensor data consistency. These priors capture the most impactful physical constraints for our application (reflectance boundedness, shape smoothness, and instrument consistency). To further evaluate the super-resolution performance, we tested it under different input configurations corresponding to the available Sentinel-2 bands at varying spatial resolutions. For instance, at 10-m resolution, only four bands (B, G, *R*, and NIR) are available. Accordingly, we assessed the model's reconstruction effectiveness using RGB bands, four-band, ten-band (available at 20-m resolution), and full 12-band inputs. We adopted peak signal-to-noise ratio (PSNR) and spectral angle similarity (SAM) for quantitative evaluation of super-resolution results, which can be calculated as follows:

$$\text{PSNR} = 10\log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right) \tag{5}$$

$$\text{SAM} = \frac{\langle S, \hat{S}\rangle}{\|S\|_2 \cdot \left\|\hat{S}\right\|_2} \tag{6}$$

where MAX = 1.0; and S and $\hat{S}$ represent the ground truth and reconstructed spectra, respectively. Higher PSNR and SAM values (with a maximum of 1) indicate better reconstruction fidelity and greater spectral similarity to the reference, respectively. Given that the model is probabilistic, we produce a 40-member ensemble of the DDPM model output.

E. Retrieving DASF and CSC

If the contribution of the soil to canopy reflectance is negligible, the canopy BRF at wavelength λ can be approximated

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TABLE III

QUANTITATIVE EVALUATION OF SUPER-RESOLUTION RESULTS. WE COMPARE THE ERRORS BETWEEN THE GENERATED BRF AND THE CORRESPONDING SIMULATED REFERENCE HYPERSPECTRAL BRF. THE TERMS MEAN, MAX, MIN, Q1, Q2, AND Q3 DENOTE THE AVERAGE, HIGHEST, LOWEST, AS WELL AS THE FIRST, SECOND, AND THIRD QUARTILES OF PSNR AND SAM

Method	PSNR						SAM					
	Mean	Max	Min	Q1	Q2	Q3	Mean	Max	Min	Q1	Q2	Q3
CNN-SSR	36.31	40.22	9.52	33.86	35.90	37.32	0.9904	0.9986	0.5523	0.9910	0.9946	0.9966
GAN-SSR	38.27	50.30	22.41	35.00	39.09	41.60	0.9957	0.9998	0.9644	0.9950	0.9975	0.9987
Physics-SSR	40.81	54.99	**28.32**	36.42	39.23	42.36	0.9988	**0.9999**	0.9676	**0.9987**	0.9991	**0.9996**
DDPM	**41.49**	**57.79**	26.13	**37.10**	**40.24**	**43.50**	**0.9989**	**0.9999**	**0.9900**	0.9985	**0.9992**	**0.9996**

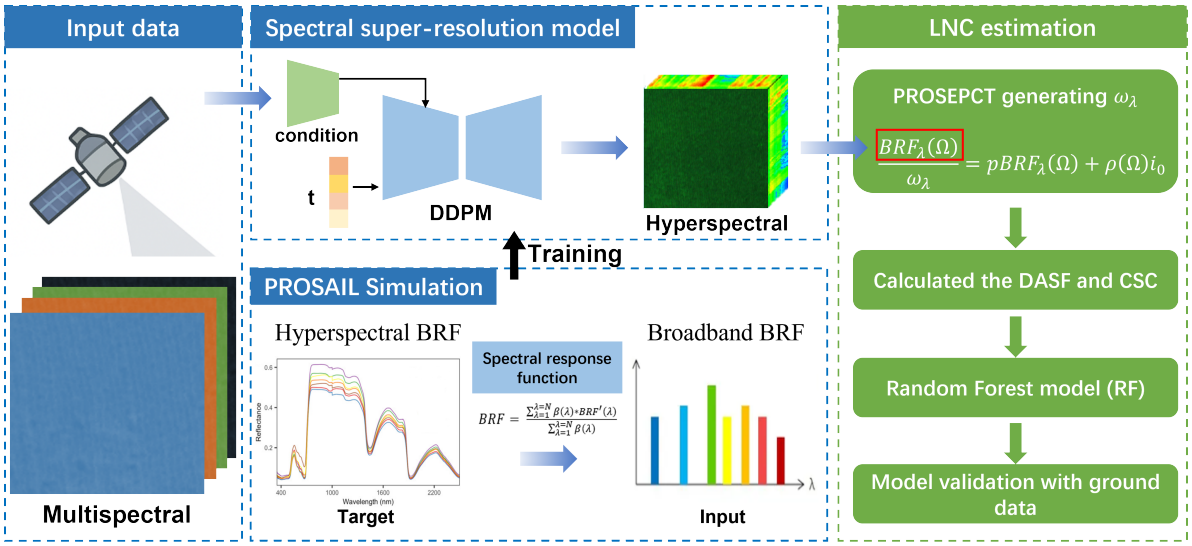


Fig. 1. Overall workflow of the proposed SSR and LNC estimation framework. Broadband multispectral or RGB input data are fed into an SSR model to reconstruct high-resolution hyperspectral reflectance. The PROSAIL model provides simulated hyperspectral–broadband BRF pairs for training and validation. The reconstructed spectra are used to calculate the DASF and CSC, which are then used in an RFR model for LNC estimation. The final results are validated against ground truth observations.

as follows [21]:

$$\mathrm{BRF}_\lambda(\Omega) = \frac{\rho(\Omega)\, i_0 \omega_\lambda}{1 - \omega_\lambda p} \tag{7}$$

where i_0 is the canopy interceptance, $\rho(\Omega)$ is the directional escape probability (also known as directional gap density), Ω is the viewing direction, p is the recollision probability, and ω_λ is the leaf albedo. The three spectral invariant parameters can be combined into DASF [21]

$$\mathrm{DASF} = \frac{\rho(\Omega)\, i_0}{1 - p} \tag{8}$$

DASF indicates part of the canopy structure contribution to BRF, and it equals the canopy BRF if leaves do not absorb radiation ($\omega_\lambda = 1$). By correcting for canopy structural effects using DASF, the CSC can be estimated based on the ratio of BRF_λ to DASF as follows [21]:

$$\mathrm{CSC} = \frac{\mathrm{BRF}_\lambda}{\mathrm{DASF}} \tag{9}$$

where $\mathrm{CSC} = (1 - p)/(1 - \omega_\lambda p)\omega_\lambda$. It represents the fraction of intercepted radiation that is reflected from or diffusely transmitted through the vegetation [47]. Therefore, the value of CSC ranges from 0 to 1.

We can derive DASF from canopy BRF spectra at 710–790 nm. The key step is to calculate the reference leaf albedo ω_λ using the PROSPECT-4 simulation by setting Nstruct, Chl, Cw, and Cm as 1.5, 16 μg/cm^2, 0.005 g/cm^2, and 0.002 g/cm^2, respectively [21]. Then, DASF can be calculated based on the relationship between $\mathrm{BRF}_\lambda/\omega_\lambda$ and BRF_λ

$$\frac{\mathrm{BRF}_\lambda(\Omega)}{\omega_\lambda} = p\mathrm{BRF}_\lambda(\Omega) + \rho(\Omega)\, i_0 \tag{10}$$

based on the slope and intercept of this linear relationship, an estimate of the DASF for the canopy can be calculated as

$$\mathrm{DASF} = \frac{\mathrm{intercept}}{1 - \mathrm{slope}} \tag{11}$$

F. Estimating LNC and Model Evaluation

After correcting for canopy structural effects using the DASF, we developed a machine learning model to estimate LNC. Random forest regression (RFR) was chosen due to its robustness to noise, ability to handle nonlinear relationships, and high performance in trait estimation tasks. RFR is an

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ensemble learning algorithm that builds multiple decision trees and outputs the average of their predictions, effectively reducing variance and mitigating overfitting. It can efficiently handle high-dimensional input data without requiring extensive feature selection [48], [49]. Moreover, it provides estimates of feature importance, offering insights into the spectral regions most relevant to the target trait. Specifically, the reconstructed full-spectrum hyperspectral reflectance was used to build an LNC estimation model, and two additional baseline inputs were constructed for comparison: 1) the original MS BRF acquired from the satellite sensor; and 2) the MS BRF normalized by NIRv (BRF/NIRv), which partially accounts for canopy structural variability

$$\mathrm{CSC}_{\mathrm{NIR}v} = \frac{\mathrm{BRF}_\lambda}{\mathrm{NIR}v} \tag{12}$$

$$\mathrm{NIR}_v = \mathrm{NDVI} \times \mathrm{NIR} \tag{13}$$

For the RFR model used to estimate LNC, we employed fivefold cross-validation to ensure robust evaluation and to obtain predicted values for all samples. For each fold, the training set consisted of 80% of the data, and the remaining 20% were used for validation. Performance metrics included the coefficient of determination (R^2), root-mean-squared error (RMSE), and the relative RMSE (RRMSE), which can be calculated as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y}_i)^2} \tag{14}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{15}$$

$$\mathrm{RRMSE} = \mathrm{RMSE}/\bar{y}_i \tag{16}$$

where y_i is the measured value, $\hat{y}_i$ is the estimated value, $\bar{y}_i$ is the average value of the measured value, and N is the number of samples.

III. Results and Discussion

A. Super-Resolution Results in the Simulated Datasets

Table III summarizes the overall reconstruction accuracy of the four super-resolution approaches—CNN-SSR, GAN-SSR, Physics-SSR, and DDPM—in terms of PSNR and SAM. Among all methods, DDPM achieved the highest mean PSNR (41.49 dB) and mean SAM (0.9989), with improvements of 0.68 dB in PSNR and 0.0001 in SAM over the suboptimal method. Its maximum PSNR (57.79 dB) and narrow interquartile span (Q1–Q3 = 37.10–43.50 dB) further demonstrate both high accuracy and strong robustness across samples. In comparison, Physics-SSR ranked second, and its higher minimum PSNR (28.32 dB) may be attributed to sensor-consistent modeling and the smoothness constraints that are more compatible with simulated data. CNN-SSR yielded the lowest average PSNR (36.31 dB) with a minimum PSNR as low as 9.52, reflecting poor consistency in the reconstruction results and occasional spectral distortion. Regarding SAM, both DDPM and Physics-SSR maintained extremely small angular deviations (average value approximately 0.9999 mean), while CNN-SSR and GAN-SSR showed noticeably lower similarity to the reference spectra.

Fig. 2 illustrates the spectral reconstruction results for four test representative samples, including the full-spectrum reflectance and the red-edge region (680–790 nm), as well as the residual error. The DDPM method demonstrates the ability to capture the general spectral shape and major absorption features, particularly in key vegetation-sensitive regions (e.g., red-edge and SWIR). It consistently preserves the spectral structure, achieves the smallest residual magnitude, and avoids the oscillations or bias seen in competing approaches. In contrast, although the other methods can reproduce the overall spectral shape correctly, they exhibit noticeably higher spectral noise. In particular, CNN-SSR tends to underestimate the reflectance in the 500–750-nm range. In the red-edge region, which is especially critical for canopy structural correction in this study, all other methods show pronounced reconstruction errors, and in some cases fail to preserve the red-edge slope.

We further compared the DASF values derived from spectra reconstructed by different super-resolution models with those computed from the original hyperspectral data (Fig. 3). Among all methods, the proposed DDPM-based SSR achieved the highest consistency with the reference DASF (R^2 = 0.975 and RMSE = 0.037), outperforming GAN-based (R^2 = 0893 and RMSE = 0.077) and physics-constrained SSR models (R^2 = 0.969 and RMSE = 0.041). The DASF value referenced by the CNN-based exhibits greater bias due to its large reflectance residual (R^2 = 0.709 and RMSE = 0.127). DASF is particularly sensitive to canopy structural variations, as it reflects the ratio of direct to diffuse scattering in the red-edge region [21]. Even small deviations in spectral shape or amplitude within the red- and near-infrared bands can cause substantial differences in DASF estimation. The superior accuracy of the DDPM model indicates its strong ability to capture nonlinear spectral–structural relationships and maintain subtle structural clues contained in the spectral data, which are often lost when using limited MS inputs.

B. Super-Resolution Performance on Ground-Based Spectrometer Data

To further validate the generalization capability of the proposed method, we conducted super-resolution experiments using ground canopy hyperspectral measurements acquired from an ASD FieldSpec4 spectroradiometer. Four samples were selected for visual comparison in Fig. 4, and the quantitative results are summarized in Table IV.

As shown in Table IV, our DDPM-based method achieves the best overall performance across all evaluation metrics. Specifically, it attains the highest average PSNR (34.24 dB) and SAM (0.9980), along with the lowest DASF estimation error (RMSE = 0.015). Physics-SSR ranks second (PSNR = 31.31 dB, SAM = 0.9967, and DASF RMSE = 0.034). GAN-SSR shows moderate spectral fidelity (PSNR = 27.55 dB and SAM = 0.9908) yet a relatively low DASF RMSE (0.025).

Visually, DDPM reconstructions successfully maintained the red-edge slope and matched the reference spectrum well in both the full-spectrum and the red-edge regions (680–790 nm), maintaining a relatively small error. Physics-SSR also exhibits

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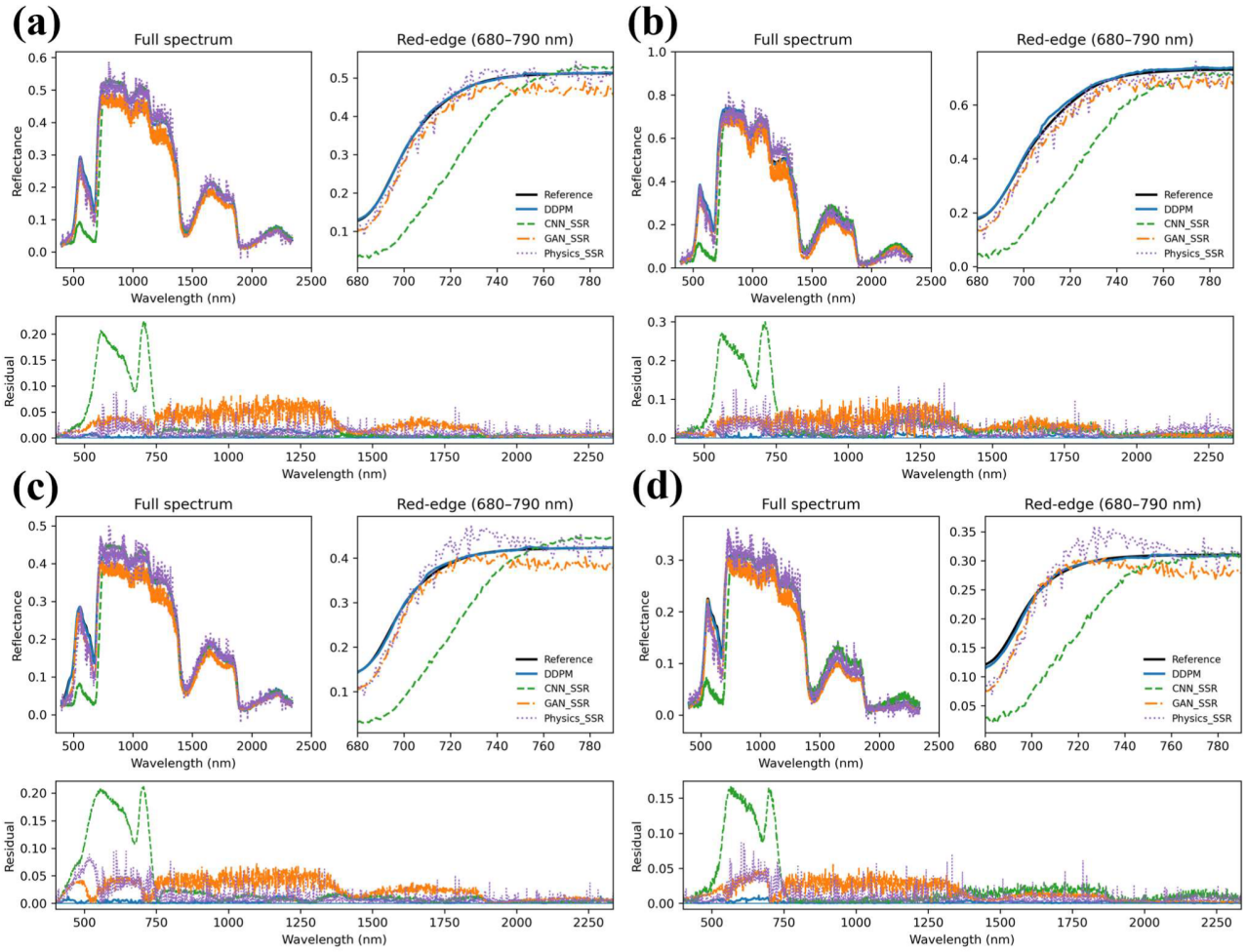


Fig. 2. Qualitative results of hyperspectral reconstruction from the simulated test dataset. (Top-left) Full-spectrum reflectance. (a)–(d) Spectral reconstruction results for four representative test samples. (Top-right) Red-edge zoom (680-790 nm). (Bottom) Residuals.

TABLE IV

QUANTITATIVE COMPARISON OF HYPERSPECTRAL SUPER-RESOLUTION PERFORMANCE ACROSS DIFFERENT METHODS IN GROUND-SYNCHRONOUS HYPERSPECTRAL VALIDATION

Method	PSNR			SAM			RMSE of DASF
	Mean	Max	Min	Mean	Max	Min	
CNN-SSR	19.97	26.29	10.32	0.9479	0.9651	0.8268	0.126
GAN-SSR	27.55	31.20	14.34	0.9908	0.9956	0.9333	0.025
Physics-SSR	31.31	37.34	15.71	0.9967	0.9989	0.9636	0.034
DDPM	34.24	43.90	16.27	0.9980	0.9998	0.9653	0.015

acceptable alignment, but significant spectral noise can be observed. GAN-SSR outputs appear more unstable and noisier, likely caused by generator instability during adversarial training. CNN-SSR tends to underestimate reflectance in the 500–750-nm region and introduces significant residual errors throughout the full-spectrum. CNN-based methods may be insufficient to capture the complex spectral variations in real-world measurements. CNN-SSR significantly overestimates reflectance in the 750–1350-nm region, underestimates reflectance in other spectral ranges, and accumulates broad residual peaks across the entire spectral range. It is worth noting that, unlike deterministic models that provide highly accurate test results in simulated settings, generative models, although slightly less accurate on simulated metrics, may have better generalization capabilities when applied to real-world data with more complex noise and structural variability [43], [50]. Overall, the DDPM method demonstrates robust performance in spectral reconstruction from real MS inputs and

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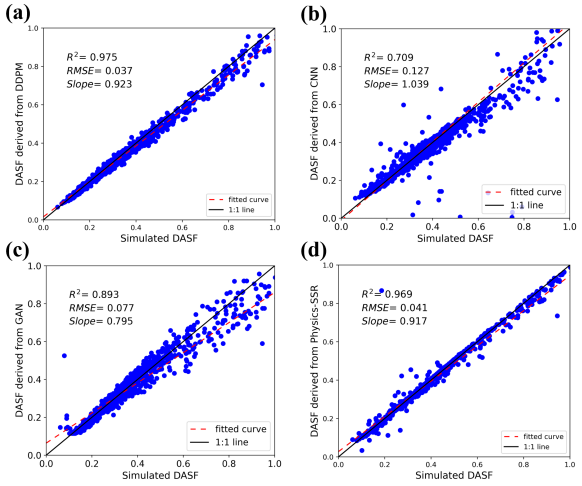


Fig. 3. Comparison between DASF values derived from reconstructed spectra using different super-resolution models. (a) DDPM-based SSR. (b) GAN-based SSR. (c) CNN-based SSR. (d) Physics-constrained SSR and reference DASF values in the test dataset.

DASF estimation accuracy. Although its performance degrades in real-world testing, the RMSE of DASF remains reliable compared to the simulated data, suggesting the model's potential for practical deployment in field-based remote sensing applications requiring precision canopy parameter retrieval.

C. Variation of DASF Across Vegetation Types

The SSR model was used for subsequent canopy correction and LNC estimation. We applied the best-performing DDPM super-resolution model to reconstruct hyperspectral BRF spectra from Sentinel-2 MS observations corresponding to ground-measured LNC samples. Based on the reconstructed hyperspectral BRF, we then computed the DASF and the CSC. The DASF exhibited clear differences across various vegetation types, as shown in Fig. 5. Among the seven vegetation types, DF plots demonstrated the highest DASF values, with maximum values exceeding 0.6. This indicates a strong directional scattering contribution from upper canopy layers, likely due to their complex crown architecture and high leaf area density. In contrast, EF showed the lowest DASF values, generally ranging from 0.20 to 0.35, suggesting reduced scattering due to differences in leaf shape and arrangement (broad, horizontal leaves versus narrow, and vertical foliage) and overall canopy density. MF yielded intermediate DASF values, typically between 0.30 and 0.45, aligning with findings from previous studies that reported transitional scattering characteristics between broadleaf and needleleaf canopies [22].

In nonforest vegetation, CRO demonstrated the highest DASF, with a median around 0.43 and relatively narrow variability, reflecting structured agricultural canopies with high light interception. WET also exhibited relatively high DASF values (median ~0.34), though with a broader spread, likely due to variability in canopy type (e.g., emergent versus low vegetation). GRA and SHR showed slightly lower median DASF values (~0.31 and ~0.33, respectively) and narrower interquartile ranges, indicating more consistent and structurally simpler canopies. Overall, broadleaf-dominated and cultivated vegetation types tend to exhibit higher DASF values, reflecting greater exposure of leaf surfaces and more complex canopy geometry. In contrast, needleleaf and low-stature vegetation such as shrubs and grasses show lower DASF values, consistent with simpler or more vertically canopy structures. These results highlight the sensitivity of DASF to differences in canopy architecture across diverse vegetation types.

D. BRF and CSC

Fig. 6 shows the average spectral curves of canopy BRF and CSC across seven vegetation types. In the original BRF spectra [Fig. 6(a)], distinct differences are observed among vegetation types, particularly in the near-infrared (700–1300 nm) and shortwave infrared (1500–2300 nm) regions. CRO and DF exhibit the highest BRF values, especially in the NIR plateau (700–1300 nm), reflecting high directional reflectance due to relatively open and horizontally structured canopies. EF displays the lowest BRF throughout the spectrum, likely due to vertically aligned, densely packed needle-like foliage that limits directional reflectance. MF, GRA, WET, and SHR fall between these extremes, with spectral differences attributable to the fraction of broadleaf species in MF, canopy height, layering, and leaf orientation [51].

After correcting for canopy structural effects using the DASF-based method, the differences among vegetation types are greatly reduced in the CSC spectra shown in Fig. 6(b). In contrast to the mean BRF, the mean CSC curves of all vegetation types converge toward similar shapes and magnitudes, particularly in the 700–1300-nm region where structural effects were previously most pronounced. This spectral homogenization suggests that much of the variation in BRF arises from structural factors rather than inherent optical differences. Notably, the broadleaf and CRO canopies, which had the highest BRF, show substantially reduced scattering coefficients after correction, while the EF curve shifts upward, highlighting its stronger intrinsic scattering per unit projected leaf area. This reversal in relative magnitude reflects the fact that BRF overrepresents structurally open canopies and underrepresents dense, vertically oriented ones. The CSC, in contrast, isolates the leaf optical contribution, removing structural bias and enabling a fair comparison of leaf biochemical trait-related scattering. Moreover, in the SWIR region, EF exhibits relatively higher CSC values compared to DF and MF. This indicates that needleleaf species may exhibit stronger intrinsic scattering in the SWIR bands, possibly due to higher dry matter content or lower internal absorption [52], [53]. Together, these results demonstrate the importance of structural correction in normalizing reflectance data and isolating intrinsic canopy optical properties. The convergence of CSC curves demonstrates that DASF correction effectively suppresses canopy architecture effects, revealing the underlying spectral properties more directly related to biochemical traits such as LNC.

We calculated the correlation between LNC and canopy BRF and CSC as shown in Fig. 7. In the original BRF-based correlation [Fig. 7(a)], strong positive correlations with LNC appear across the visible- and near-infrared

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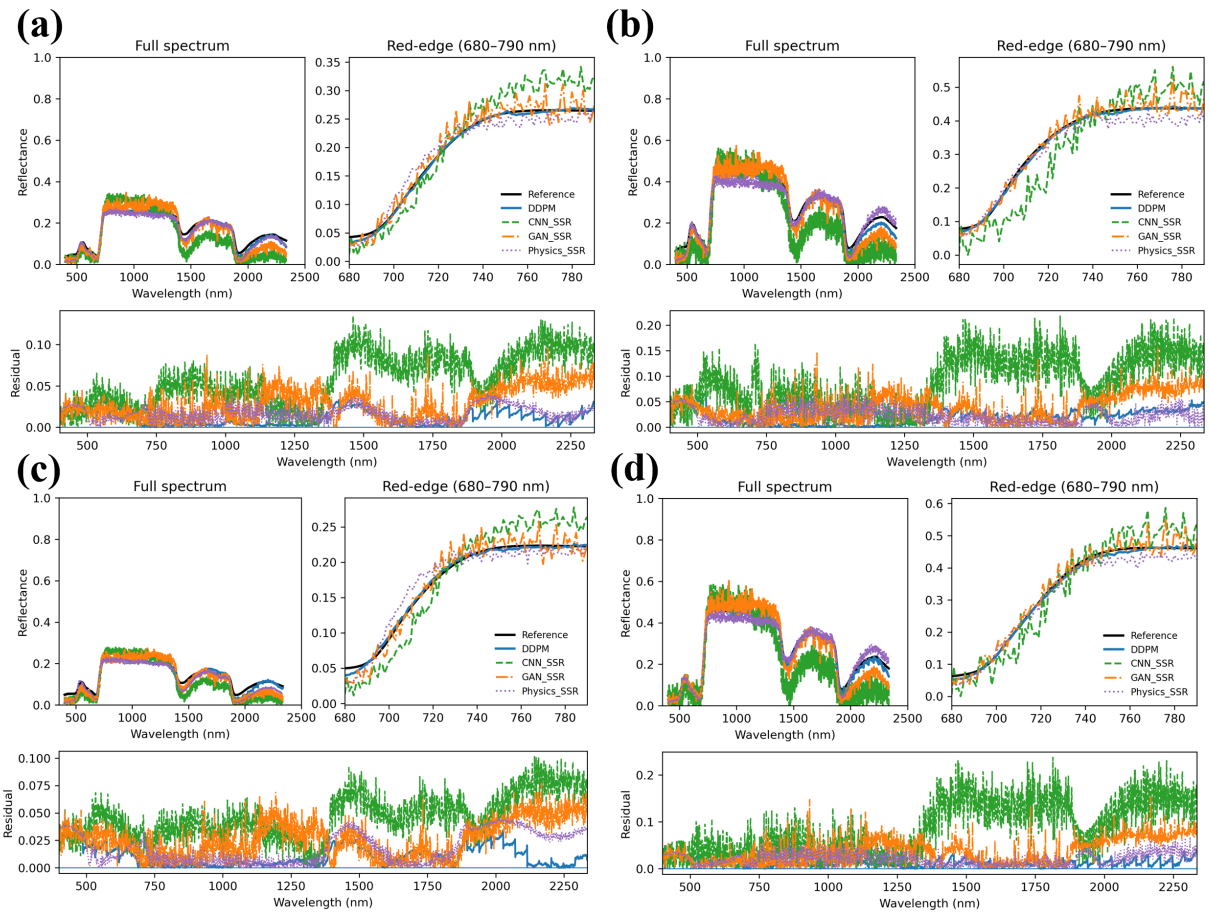


Fig. 4. Qualitative results of hyperspectral reconstruction from the groundsynchronous hyperspectral validation dataset. (a)–(d) Spectral reconstruction results for four representative test samples. (Top-left) Full-spectrum reflectance. (Top-right) Red-edge zoom (680-790 nm). (Bottom) Residuals.

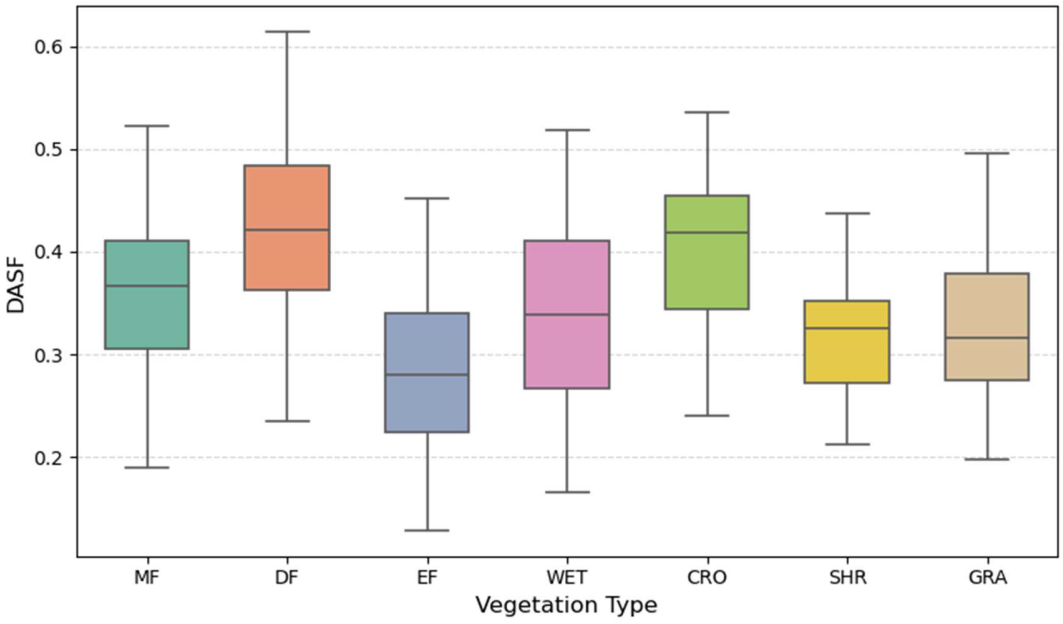


Fig. 5. Boxplot of the DASF retrieved across seven vegetation types: MF, DF, EF, WET, CRO, SHR, and GRA.

regions, particularly in the red-edge (~700–740 nm) and NIR plateau (~750–1300 nm). This trend is consistent with the fact that nitrogen is closely associated with chlorophyll content, which governs absorption in the visible range, and also affects internal leaf structure that modulates NIR scattering [54], [55]. However, these correlations may be confounded by

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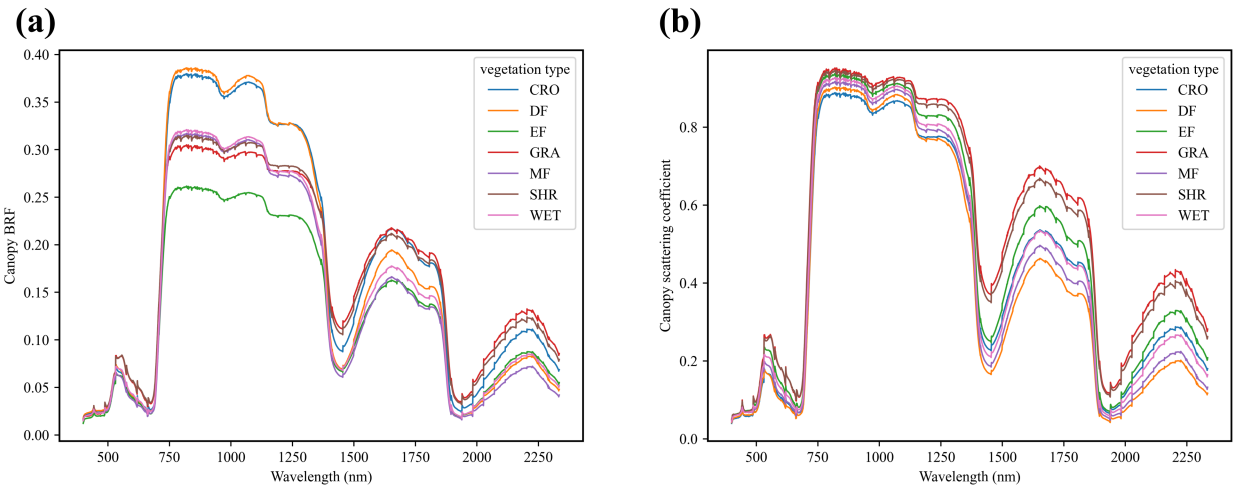


Fig. 6. (a) Original canopy BRF spectra before correction. (b) CSC spectra averaged by vegetation type after correcting for canopy structural effects using the DASF method. Before correction, significant spectral differences are observed among vegetation types. After correction, these differences are markedly reduced, suggesting that structural effects are the dominant source of reflectance variation across canopy types.

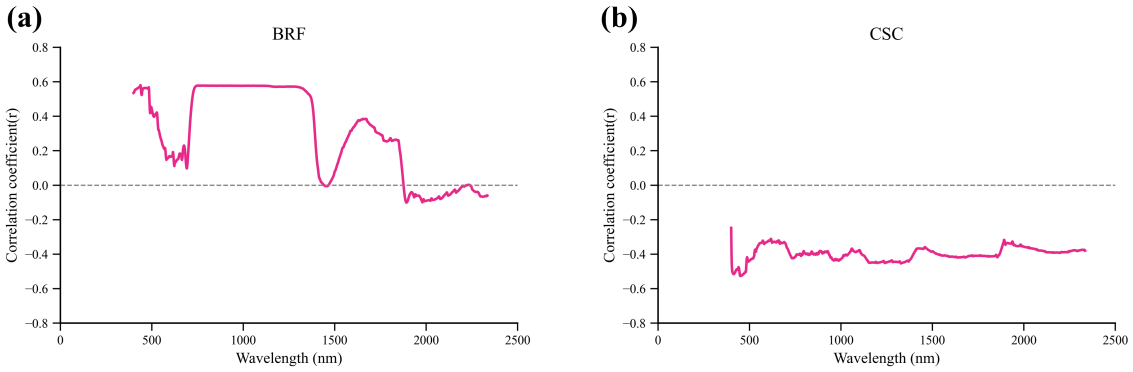


Fig. 7. Correlation coefficient (r) between LNC and the canopy. (a) BRF and (b) CSC.

canopy structural effects, which make the correlations in the NIR positive regardless of the biochemical content. Notably, a clear positive correlation is also seen around 1750 nm, corresponding to a known water absorption feature, which may reflect covariation between leaf nitrogen and water content under natural conditions. In contrast, the correlation between LNC and the DASF-corrected CSC is fundamentally altered [Fig. 7(b)]. The previously strong positive correlations in the NIR region now shift to negative values, particularly beyond 800 nm. This shift aligns with previous findings [21], where the intrinsic scattering properties of leaves—after accounting for canopy structure—are negatively associated with nitrogen content, due to increased absorption in nitrogen-rich leaves. Interestingly, in the strong water absorption region beyond 1900 nm, the original BRF showed negligible correlation with LNC, likely due to canopy-level scattering and noise masking any physiological signal. After DASF correction, the CSC revealed a moderate negative correlation in this region, suggesting that higher nitrogen content—potentially associated with increased protein or dry matter—enhances absorption even in these water-dominated wavelengths.

E. Performance of Successive Input-Correction Strategies for LNC Retrieval

After reconstructing the hyperspectral BRF and canopy correction, we perform leaf nitrogen estimation. To further assess how different super-resolution and structural correction strategies influence LNC estimation, we compared the regression results across multiple input types, including original MS data, hyperspectral reconstructions from DDPM-, CNN-, GAN-, and physics-based SSR models, as well as their corresponding CSC. The results are shown in Fig. 8. Sentinel-2 as input provides a baseline with an R^2 of 0.445, RMSE of 0.495%, and RRMSE of 26.93% [Fig. 8(a)]. While hyperspectral data provide richer spectral information, they also contain considerable redundancy, especially with narrow bands across broad wavelength ranges. The results show that accurately reconstructing the hyperspectral enables improved LNC estimation accuracy. Among the hyperspectral reconstructions, the DDPM-based super-resolution yielded the highest retrieval accuracy (R^2 = 0.469, RMSE = 0.484, and RRMSE = 26.37%). The physics-constrained SSR also achieved relatively high accuracy (R^2 = 0.459 and RMSE = 0.489) but still slightly lagged behind

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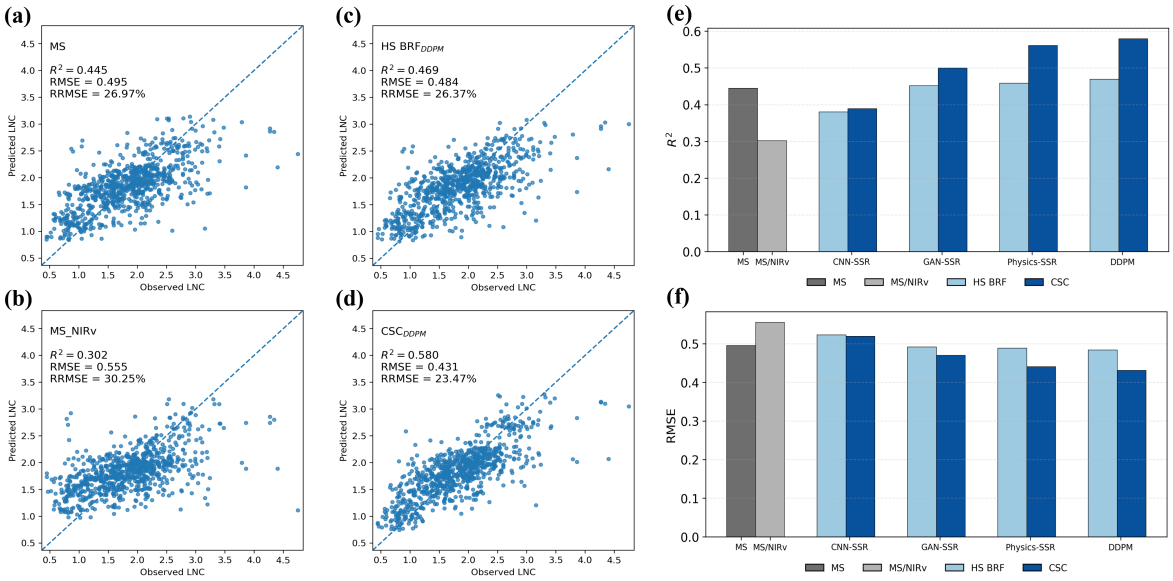


Fig. 8. Retrieval accuracy of LNC using the RF model with the input. (a) MS, (b) MS/NIRv, (c) hyperspectral BRF reconstructed by DDPM, (d) CSCs derived by BRF and DASF, (e) R^2, and (f) RMSE before and after correction by different models.

the DDPM model (Fig. 8). This advantage aligns with the spectral evaluation results, confirming that the diffusion-based generative model can more effectively capture fine-grained spectral variations relevant to canopy biochemical properties. In contrast, the CNN-SSR exhibited a clear degradation in performance (R^2 = 0.380 and RMSE = 0.523), performing even worse than the MS baseline. When structural effects are further corrected using DASF-derived CSC, the predictive performance improves notably. The CSC of the DDPM model achieved the best overall performance (R^2 = 0.579, RMSE = 0.431, and RRMSE = 23.47%), improving the estimation of LNC, with the RMSE values reduced by 14.8% compared to MS. The CSC of the Physics-SSR model (R^2 = 0.561 and RMSE = 0.440) also performed well, likely due to the spectral smoothness of the physics-constrained reconstruction. Compared with uncorrected hyperspectral inputs, CSC reduced estimation errors by 4.5%–12.3%, demonstrating its effectiveness in normalizing canopy scattering variability that otherwise confounds nitrogen retrieval. However, when the reconstructed hyperspectral BRF has large reconstruction errors, these errors cannot be compensated for by subsequent correction methods. For example, the CSC of the CNN-SSR model only achieved (R^2 = 0.389 and RMSE = 0.519), resulting in even greater errors in LNC estimation (Fig. 8).

We found that the NIRv-based correction (MS/NIRv) underperformed significantly (R^2 = 0.302 and RMSE = 0.555) [Fig. 8(b)], indicating limited capability to represent canopy structure from MS information alone. We compared the DASF value calculated based on DDPM with NIRv and found a significant error, as shown in Fig. 9. Although the two sets of values are positively correlated, almost all NIRV values are lower than DASF values. Therefore, we believe that the poor performance can be attributed to several limitations of NIRv when used for LNC estimation. First, NIRv uses only red and NIR bands, insufficient to represent the complex

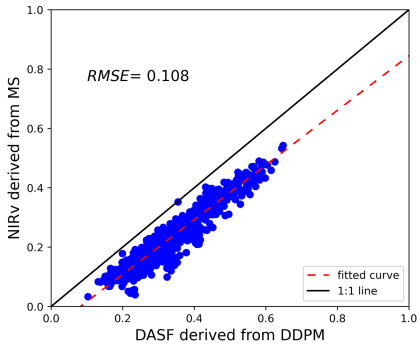


Fig. 9. Relationship between DASF derived from DDPM-reconstructed hyperspectral reflectance and NIRv computed from MS data.

interactions between multiple scattering, leaf angle distribution, and biochemical absorption that jointly influence canopy reflectance. Moreover, NIRv inherits the nonlinear saturation behavior of NDVI under dense canopy conditions, leading to suppressed sensitivity to structural variation. Third, because our analysis relies on real airborne MS measurements, NIRv is also affected by sensor noise and spectral mixing. These mixed signals amplify errors when NIRv is used as a proxy for DASF. As a result, NIRv-based correction cannot fully compensate for variations in canopy architecture or nitrogen-induced spectral shifts. The effectiveness of NIRv in reducing the impact of canopy structure on BRF-based LNC estimation may still require further validation across large-scale datasets covering diverse sites and plant species.

Fig. 10 illustrates LNC regression performance based on the DDPM method in various vegetation types. We found that CSC consistently achieves the lowest RMSE across all vegetation types, with RMSE reduced by 0.026–0.21 compared with MS. The HS yields moderate improvements over MS but

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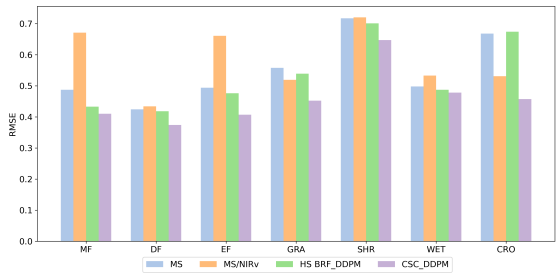


Fig. 10. RMSE of LNC estimation models with different inputs across vegetation types.

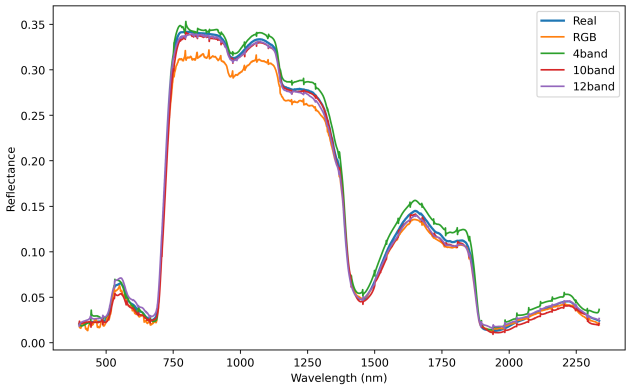


Fig. 11. Comparison of the average hyperspectral reflectance reconstruction results under different input spectral configurations for the DDPM model. Inputs include RGB bands (B2–B4), four-band (B2–B4 and B8), ten-band (B2–B8, B8a, B11, and B12), and full 12-band Sentinel-2 multispectral inputs. The real hyperspectral reflectance is shown in blue for reference. The model shows improved reconstruction accuracy with increasing input band number.

is inferior to CSC. These findings demonstrate that DASF provides a physically grounded normalization that reduces structural confounding and improves cross-type robustness. Forest classes (MF, DF, and EF) exhibit relatively low RMSE for all methods. In contrast, SHR and GRA show higher overall errors, likely due to stronger soil or background influence, which likely limits model accuracy even after spectral reconstruction and DASF correction. Notably, the MS/NIRv showed the greatest performance degradation in MF and EF, with an increase in RMSE of 0.16–0.18, while the DASF-based method maintained a lower RMSE, further demonstrating that DASF is more robust than NIRv in eliminating structural variability.

F. Hyperspectral Super-Resolution and LNC Estimation Under Limited Spectral Inputs

To evaluate the robustness of super-resolution models under varying spectral constraints, we compared the reconstruction results using the DDPM model with different subsets of Sentinel-2 bands, including RGB (B2–B4), four 10-m bands (B2–B4 and B8), and ten 10–20-m bands (B2–B8, B8a, B11, and B12). Quantitatively, PSNR and SAM metrics improved steadily with more input bands. The DDPM model's PSNR increased from 30.32 (RGB) to 40.31 (ten bands). DASF estimation accuracy, measured by R^2 against the reference value, also exhibited a significant increase with band count. The R^2 improved from 0.370 to 0.977 (Table V).

TABLE V
PERFORMANCE METRICS OF DDPM UNDER DIFFERENT INPUT BAND CONFIGURATIONS. METRICS INCLUDE THE MEAN PSNR, SAM, AND R^2 AND RMSE OF DERIVED DASF

Input	DDPM		
	PSNR	SAM	R²/RMSE
3bands	30.32	0.9912	0.370/0.18
4bands	33.82	0.9926	0.903/0.072
10bands	40.31	0.9988	0.956/0.049
12bands	**41.49**	**0.9989**	**0.977/0.035**

When using only RGB bands (B2–B4), the DDPM model struggles to recover the true hyperspectral shape—particularly in the red-edge (~700–750 nm), NIR plateau (~800–1300 nm), and SWIR absorption features (~1400 and ~1900 nm), reflecting limited capacity to reconstruct detailed spectral variability with such limited input (Fig. 11). This stems from the fact that RGB bands provide limited information on the canopy's internal scattering and absorption features, especially in regions sensitive to leaf biochemical traits (e.g., red-edge, NIR, and SWIR). However, introducing even a single additional band in the NIR region (e.g., B8) substantially improves model stability and DASF reliability. NIR reflectance is strongly influenced by internal leaf and canopy structure and is thus crucial for differentiating between vegetation types and estimating scattering-related indicators [21]. The sharp improvement in DASF R^2 demonstrates the critical role of NIR information in constraining spectral generation and preserving physically meaningful features.

We evaluated the LNC estimation performance under different input configurations (Table VI). Under RGB-only input, all methods showed significant performance degradation. For DDPM, the hyperspectral reconstruction yielded a relatively low LNC R^2 (0.3 for BRF and 0.34 for CSC). Interestingly, while its performance is still not as good, it is comparable to the original MS model using four bands with the addition of the NIR band (R^2 = 0.33). This suggests that even without NIR input, the framework of SSR combined with canopy structure correction can effectively leverage spatial–spectral priors learned during training to reconstruct relevant spectral information that supports nitrogen estimation. Adding the NIR band (B8) led to a substantial jump in LNC estimation accuracy, with R^2 values of 0.43 (BRF) and 0.45 (CSC) based on the hyperspectral output of DDPM. This confirms the crucial role of NIR reflectance in characterizing vegetation biochemical traits related to absorptions. As more bands are added, the estimation accuracy improves further, but with diminishing returns. As for the CSC derived using NIRv, it still performs poorly under different band inputs.

The results showed that the DDPM model, particularly when supported by minimal NIR information, can reliably reconstruct hyperspectral reflectance sufficient for biophysical

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TABLE VI
R^2 RESULTS OF LNC ESTIMATION UNDER DIFFERENT INPUT SPECTRAL CONFIGURATIONS AND MODEL-CORRECTION STRATEGIES

Input	MS		DDPM	
	MS	MS/NIR_V	BRF	CSC
3bands	0.20	/	0.30	**0.34**
4bands	0.33	0.19	0.43	**0.44**
10bands	0.42	0.29	0.47	**0.53**
12bands	0.44	0.30	0.47	**0.58**

trait estimation, such as LNC. Also, the proposed super-resolution models can still achieve significant improvements in trait estimation even with extremely limited input (e.g., RGB-only). It reinforces the potential of combining SSR and structural normalization in applications with limited spectral availability, such as RGB sensors on UAVs or where the sampling resolution results in fewer available bands.

G. Limitations and Future Work

The proposed framework leverages DASF-based structural normalization on DDPM-reconstructed hyperspectral reflectance to improve LNC estimation, but its applicability is subject to several important limitations. First, the DASF formulation assumes that the measured signal is dominated by multiple scattering within a dense and horizontally homogeneous canopy [21]. Under such conditions, variations in reflectance can be effectively decomposed into a structural component (described by DASF) and a biochemical component. In practice, many landscapes—particularly GRAs, SHRs, and early season CROs—exhibit substantial bare soil or understory exposure. Soil spectra are typically brighter and spectrally smoother than vegetation and can dominate the red and NIR bands at low LAI, causing DASF to absorb soil-induced variability rather than pure canopy scattering [56], [57]. This contamination may lead to biased CSC values and partly explains the weaker performance observed for GRA and SHR classes in our type-specific analysis (Section III-E), where canopy–soil mixing is strong. Second, the applicability of the proposed method across seasons remains limited by the fact that canopy structural and leaf optical properties exhibit substantial intra-annual variability across most vegetation types. LAI, leaf angle distribution, leaf lifespans, and water content undergo pronounced seasonal fluctuations in both natural ecosystems and agricultural systems, which may affect the effectiveness of spectral reconstruction and CSC correction [58]. For example, strong seasonal cycles in leaf area and canopy scattering behavior in Congolese rainforests, driven largely by rainfall and radiation regimes [59]. The seasonal dimension was not explicitly tested in this study. Third, deviations of different vegetation types are mainly influenced by differences in leaf morphology and canopy architecture, as well as phenological variations and soil background, as mentioned above. Although the proposed framework has been evaluated across multiple vegetation types and shows generally consistent performance, a more comprehensive evaluation would require incorporating more representative vegetation types due to uneven sample size distribution across types.

Finally, the super-resolution models were trained exclusively using simulated hyperspectral data generated by the PROSAIL model, without incorporating actual canopy reflectance measurements. Such synthetic data may not fully represent the complexities encountered in real-world scenarios, potentially limiting the generalization capability of the super-resolution models [60], [61]. Although the PROSAIL parameter ranges and sampling strategy were chosen to span a broad space, they still represent an idealized canopy–soil system and do not explicitly encode all sources of variability [47]. We have not yet verified the SSR performance outside the rice canopy reflectance. Consequently, both the reconstructed hyperspectral and the derived CSC may be less accurate when applied to other vegetation types and complex scenarios, such as extremely bright soils, very sparse canopies, or complex multilayer forests. Therefore, to further calibrate and evaluate the model performance, it is essential to incorporate real airborne or field-collected hyperspectral reflectance data into the super-resolution models' training process. Future work should collect more ground samples, encompassing various vegetation types, canopy structural characteristics, and seasonal variations to quantify seasonal and soil background sensitivity of DASF and CSC, and perform vegetation type-specific calibrations to enhance its generalizability.

In our study, when only RGB bands were available, the DDPM model struggled to reconstruct accurate hyperspectral spectra, leading to significantly degraded LNC inversion performance. This implies that erroneous or poorly constrained hyperspectral reconstructions may not only fail to improve, but can actually hinder, downstream trait retrieval—particularly when key spectral regions such as the NIR or red-edge are absent. In contrast, MS data are directly measured and thus typically more radiometrically stable. Especially, when key bands (e.g., NIR) are included, MS indices or NIRv-based corrections can still yield robust estimations of canopy biochemical properties (e.g., Chl). Nevertheless, as demonstrated

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by our results, hyperspectral super-resolution methods become advantageous once minimal critical bands are provided, such as adding the NIR band to the RGB configuration. In such cases, models like DDPM can leverage learned spectral priors to recover useful high-dimensional reflectance features and enable effective structural correction via CSC. This highlights the potential of combining accessible MS data with learning-based hyperspectral enhancement to approximate the benefits of full hyperspectral sensing, albeit with caution to ensure spectral reliability. Future work should therefore carefully assess the reliability and generalizability of hyperspectral reconstructions under varying input conditions, and explore hybrid frameworks that dynamically adapt between MS and hyperspectral pathways based on data quality and availability.

IV. Conclusion

This study presents an integrated framework for estimating LNC by leveraging super-resolved hyperspectral reflectance and structure-normalized CSC. The DDPM was trained on simulated canopy spectra to reconstruct full-range hyperspectral reflectance from Sentinel-2 MS inputs. These reconstructions were then used to compute DASF and derive CSC, enabling structure-normalized LNC estimation based on spectral invariance theory even under spectral limitations. Our results show that the DDPM model not only achieved superior spectral reconstruction fidelity compared to baseline models (CNN, GAN, and physics-constrained) but also outperformed them in downstream LNC estimation tasks. The DASF-corrected CSC effectively eliminated canopy structural effects, leading to significant improvements in estimation accuracy across diverse vegetation types and observation conditions. The framework maintains good performance when different available bands are input, and adding just one NIR band can significantly improve reconstruction quality and subsequent estimation accuracy. Overall, this frame work demonstrates robust performance under both simulated and real-world conditions, offering a practical solution for accurate, scalable, and structure-aware LNC estimation from commonly available satellite imagery.

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